

# Skill Trends in East Asia

Insights from Online Job Postings

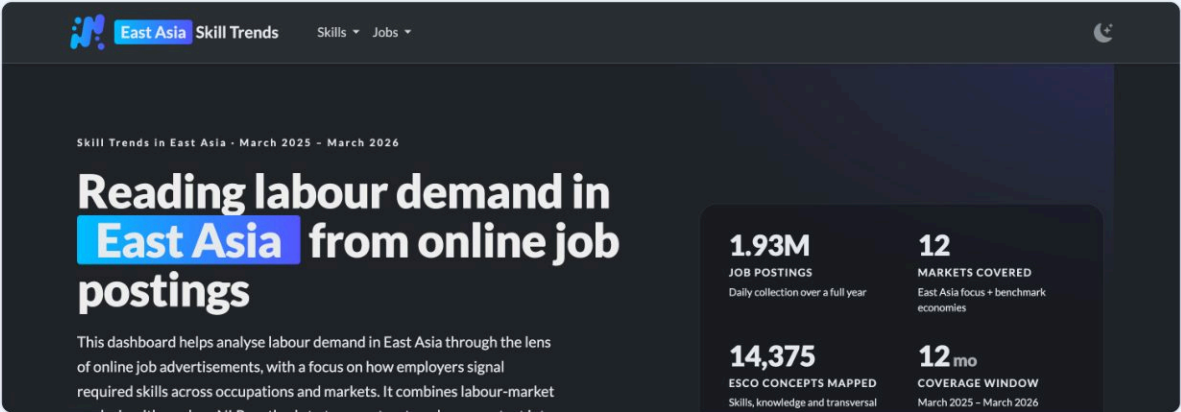
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**This is a static version of the interactive report.**

The fully interactive version, with switchable charts and downloadable data, is available at [skillmap-nlp.github.io/east-asia/report.html](https://skillmap-nlp.github.io/east-asia/report.html). In the static version, charts that can normally be toggled are shown in their default state.

In addition to this report, an interactive dashboard is available at [skiltrends.pythonanywhere.com](https://skiltrends.pythonanywhere.com).

The underlying data for the report can be downloaded from [this Google Drive folder](#).



Preview of the interactive dashboard.

## Executive Summary

This report analyses labour demand in East Asia through the lens of online job advertisements, with a focus on how employers signal required skills across occupations and markets. It combines labour-market analysis with modern NLP methods, including automated translation, sentence-level classification, and LLM-assisted skill extraction, to turn unstructured vacancy text into comparable measures of skill demand.

Using ESCO-based mappings, the report compares the composition of requested skills and knowledge areas across countries, with special focus on digital competencies and indicators of relative labour demand derived from job posting intensity. Labour markets across East Asia are undergoing rapid transformation, with increasing digitization reshaping how employers search for talent. Online job postings (OJPs) provide a useful proxy for real-time labour demand, particularly for mid- and high-skill occupations.

|                      |                                                   |
|----------------------|---------------------------------------------------|
| JOB ADS<br>1,982,965 | MARKETS ANALYZED<br>9 East Asian<br>+ 3 benchmark |
| ESCO SKILLS<br>13.9K | DATA PERIOD<br>Mar 2025 — Apr 2026                |

## Regional Coverage

### Countries covered

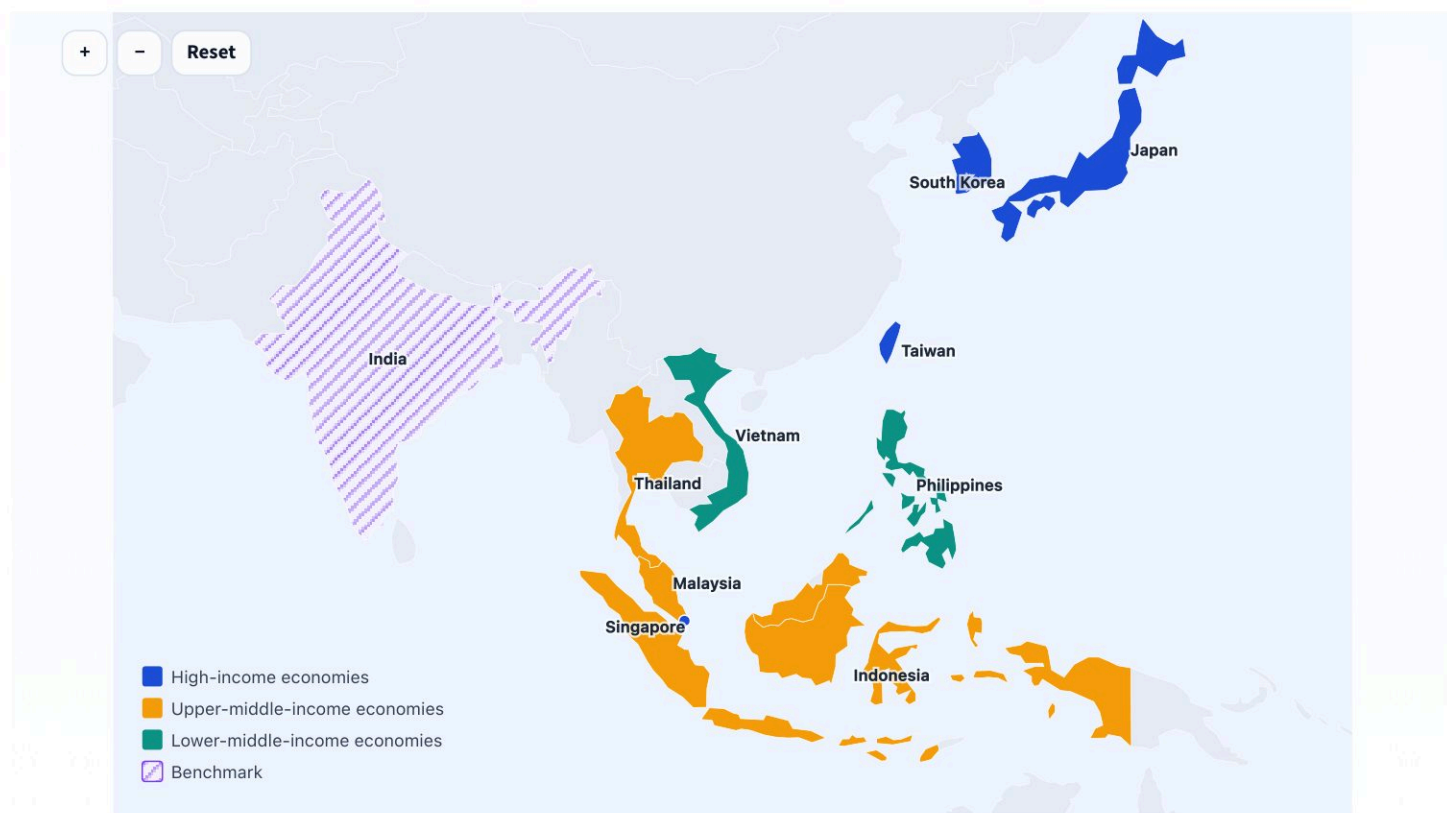


Figure 1: Analyzed countries grouped by income status.

Lower-middle-income: Vietnam, Philippines

Upper-middle-income: Thailand, Indonesia, Malaysia

High-income: Japan, South Korea, Singapore, Taiwan

Benchmark: India, Mexico, Poland

## Relative labour demand

The table below summarizes the absolute number of job offers and the number of offers per 100,000 people in the labor force. This allows us to compare both the scale of observed hiring and its intensity relative to each country's labour supply. Click a column header to sort the table.

| Country     | Income group ↑      | Job offers | Offers per 100k labor force | Real GDP growth (2025) |
|-------------|---------------------|------------|-----------------------------|------------------------|
| Japan       | High-income         | 302,262    | 435.6                       | 0.1%                   |
| South Korea | High-income         | 169,933    | 569.3                       | 2.0%                   |
| Singapore   | High-income         | 149,582    | 4003.5                      | 4.4%                   |
| Taiwan      | High-income         | 117,672    | 978.3                       | 8.6%                   |
| Malaysia    | Upper-middle-income | 176,662    | 943.7                       | 5.1%                   |
| Indonesia   | Upper-middle-income | 113,505    | 77.2                        | 5.0%                   |
| Thailand    | Upper-middle-income | 65,544     | 160.4                       | 2.5%                   |
| Philippines | Lower-middle-income | 175,615    | 336.4                       | 5.7%                   |
| Vietnam     | Lower-middle-income | 134,424    | 235.7                       | 7.1%                   |
| India       | Benchmark           | 223,220    | 36.1                        | 6.5%                   |
| Mexico      | Benchmark           | 219,762    | 356.1                       | 1.4%                   |
| Poland      | Benchmark           | 134,784    | 748.1                       | 3.0%                   |

### MACROECONOMIC SOURCES

[World Bank labor force \(2025\)](#) ↗

[World Bank real GDP growth \(2025\)](#) ↗

[National Statistics, Republic of China \(Taiwan, 2025\)](#) ↗

## Data and Methodology

### Data collection

We collect job advertisements from **Indeed** and **Careerjet**, two large aggregators with country-specific sites across the markets covered in this study. The analytical sample shown here comprises **1,982,965 postings** covering March 2025 – April 2026. Collection ran **daily for a full year** because high-volume portals often stop returning results after a limited number of pages. Collecting every day reduced the risk of missing vacancies that would no longer be reachable if the portals were queried less frequently.

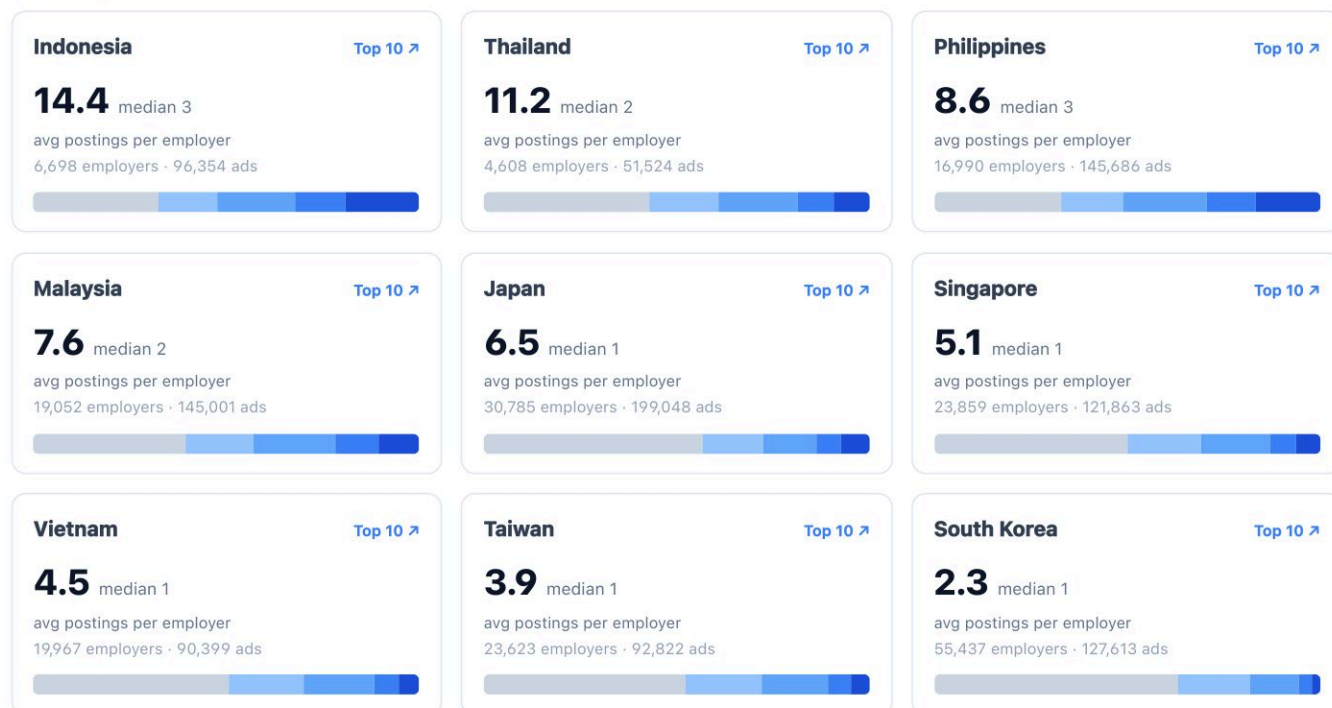
[Read more ↗](#)

## Postings per employer

Online vacancies are not evenly spread across firms: in every market a small number of high-volume recruiters and large employers account for a disproportionate share of postings, while most companies appear only once or twice. The cards below show the **average number of postings per employer**, with the median shown alongside it as a robust reference point, together with the distribution of employer posting volumes in each country. Click a country to see its **top 10 employers by posting volume**.

Employers by posting volume: 1 posting 2 3–5 6–10 11+

### EAST ASIA



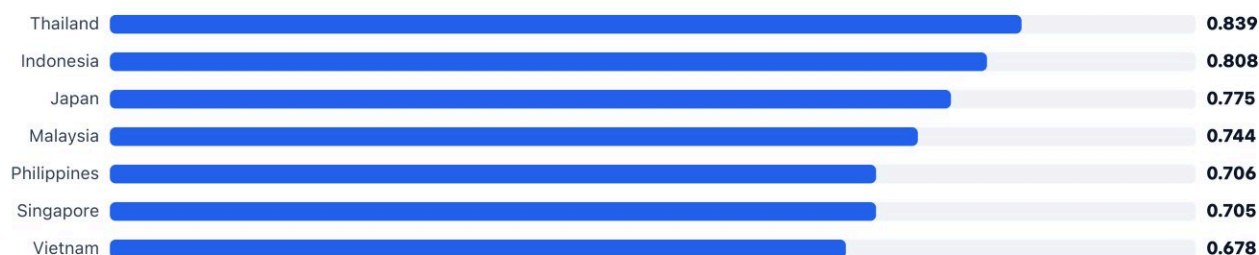
### BENCHMARK

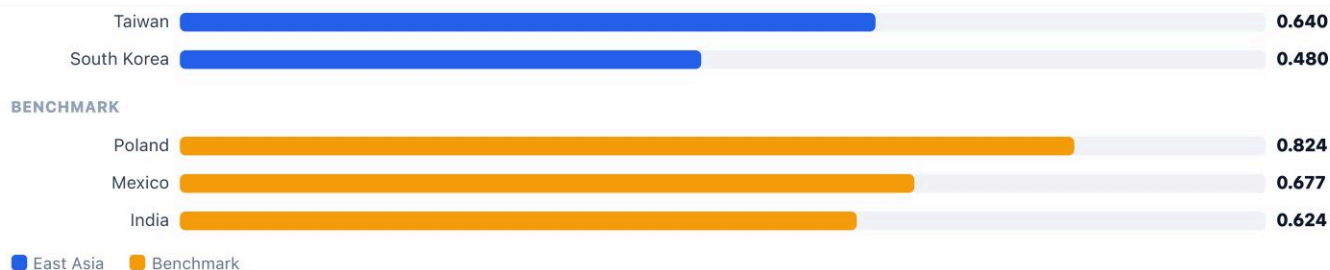


## Concentration of postings across employers (Gini)

Online vacancy postings are highly concentrated across employers in every market, with Gini values ranging from 0.48 to 0.84 — meaning a small number of recruiters account for a disproportionate share of advertised vacancies everywhere. Concentration is highest in Thailand, Poland and Indonesia, and notably lower in South Korea. These figures capture the concentration of online posting activity rather than underlying labour-market structure, and elevated values are commonly driven by recruitment agencies and job aggregators that post at volume on behalf of multiple employers.

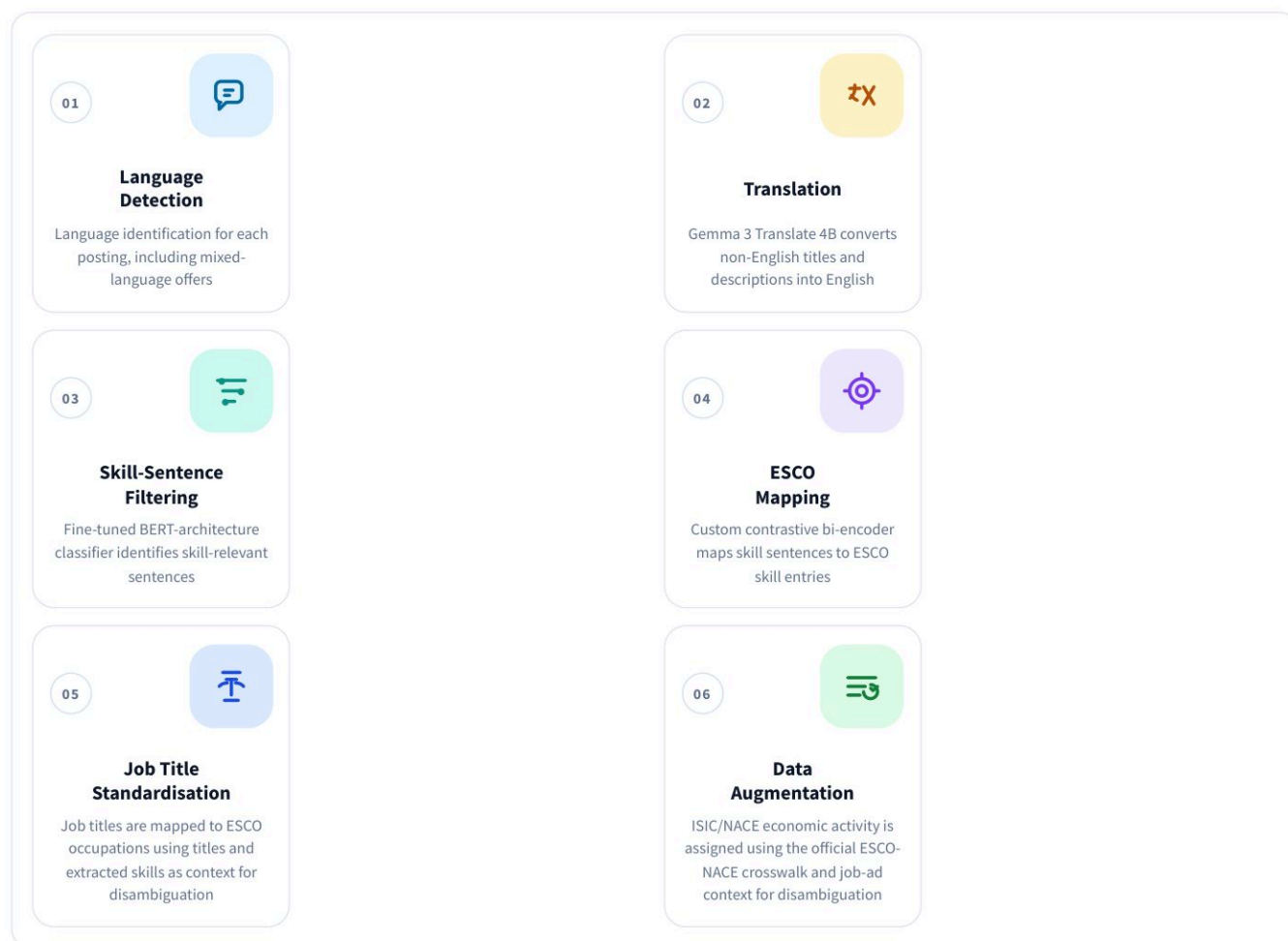
### EAST ASIA





## Processing pipeline

Raw job postings arrive in different languages and contain noisy, unstructured information. We transform them into a standardised analytic dataset in six steps: we first identify the posting language, translate non-English text into a common language, isolate skill-relevant sentences, map extracted skills to ESCO, standardise job titles to ESCO occupations using both titles and skill context for disambiguation, and then enrich each posting with sector information by assigning ISIC/NACE economic activity codes.



The detailed technical description of each processing step is provided in the Technical Annex.

## Results

### Online demand vs. actual employment structure

The table below contrasts the occupational composition of online job postings (**demand**) with actual employment shares from national labour force surveys reported to the ILO (**employment**), both at the ISCO-08 Major Group level (1-digit). Armed forces are excluded from both the online-posting and official-employment distributions. For Taiwan, the employment benchmark comes from the official DGBAS Manpower Survey rather than ILOSTAT; it uses Taiwan's ROC occupational classification, which is based on ISCO-08 but not ILOSTAT-harmonised.

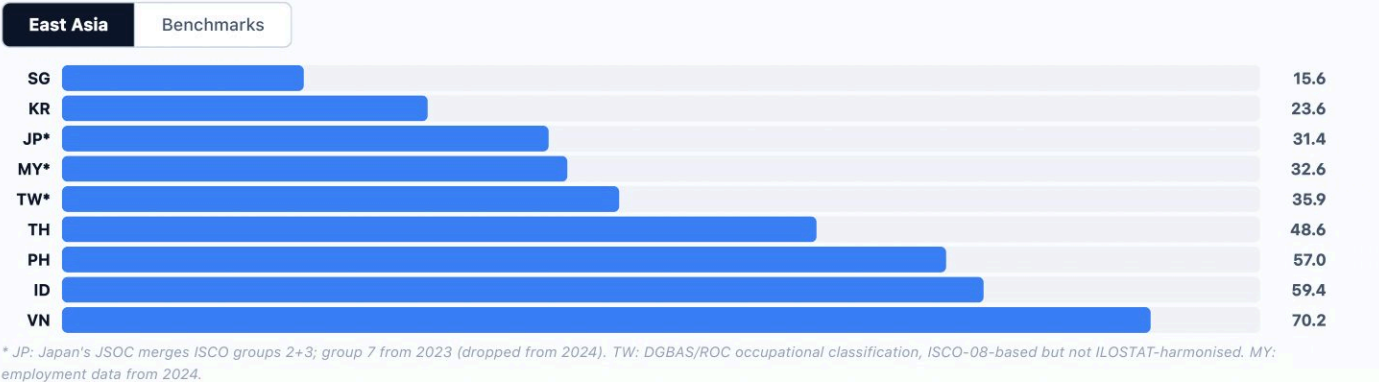
For each country we compute the **Index of Dissimilarity**, originally developed to measure residential segregation by race and gender. The same functional form is widely applied in labour market mismatch research — for instance, [Beręsewicz et al. \(2024\)](#) uses it to quantify educational and occupational mismatch between online job postings and employer-survey data in Poland.

Mismatch<sub>j</sub> = 1/2 ∑<sub>i</sub> |D<sub>ij</sub> - E<sub>ij</sub>|

where  $D_{ij}$  is the share of online job postings in ISCO group  $i$  for country  $j$  and  $E_{ij}$  is the corresponding employment share from national labour force surveys. The index expresses the percentage of the employment structure that would need to shift between occupational groups to match the composition of online demand.

Use the country selector below to browse the detailed breakdown. The summary bar chart ranks all countries by mismatch.

Index of Dissimilarity by country (pp)



Detailed ISCO breakdown by country

East Asia

Benchmarks

Country: Japan (JP) \* — empl. 2023

31.4 pp mismatch

| ISCO-08 Major Group             | Demand (%) | Empl. (%) 2023 | Gap (pp) |
|---------------------------------|------------|----------------|----------|
| 1 Managers                      | 8.0        | 1.9            | +6.1     |
| 2+3 Professionals & Technicians | 41.8       | 19.4           | +22.4    |
| 4 Clerical support              | 4.8        | 21.2           | -16.4    |
| 5 Service & sales               | 18.8       | 26.6           | -7.8     |
| 6 Skilled agriculture           | 0.3        | 2.9            | -2.6     |
| 7 Craft & trades                | 6.9        | 9.3            | -2.4     |
| 8 Plant & machine operators     | 10.1       | 7.2            | +2.9     |
| 9 Elementary occupations        | 9.3        | 11.5           | -2.2     |

Note: Japan's Standard Occupational Classification (JSOC) merges ISCO groups 2 and 3 into one category. Groups 2+3 are shown combined for comparability. Group 7 last reported in 2023 (merged into 8 from 2024). Mismatch index computed on comparable groups.

Demand = share of online job postings (this study, 03.2025–04.2026). Employment = share of civilian employment from national labour force surveys reported to [ILOSTAT](#) (indicator EMP\_TEMP\_SEX\_OCU\_NB, latest available year per country), except Taiwan from DGBAS Table 52. Gap = demand – employment (pp). Green bars indicate demand exceeds employment; red bars indicate employment exceeds demand. Armed forces excluded from both distributions. Taiwan uses ROC occupational classification, ISCO-08-based but not ILOSTAT-harmonised. Malaysia: employment data from 2024. Japan: JSOC merges ISCO 2+3; group 7 from 2023 (dropped from 2024). Mismatch computed on comparable groups.

Skill content of job postings

Skills per offer

The number of identifiable skills per posting varies substantially across countries, reflecting differences not only in underlying labour demand but also in how online vacancies are written and used. In some markets, postings resemble detailed job descriptions and explicitly list technical, language, and transversal requirements, often in more standardized HR formats; in others, they are shorter and function primarily as vacancy notices, with much of the screening and clarification taking place later in the recruitment process. Cross-country differences may also partly reflect variation in the occupational composition of online demand, as professional and technical roles tend to generate more skill-rich vacancy texts than more generic or less formal postings. The indicator should therefore be interpreted primarily as capturing the explicitness and codification of skill demand in online recruitment, rather than as a direct measure of job complexity across countries.

The table below reports the distribution of ESCO-mapped skills per posting. Click a column header to sort.

| Country | Mean ↓ | P25 | Median | P75  | P90  |
|---------|--------|-----|--------|------|------|
| Poland  | 10.0   | 6.0 | 10.0   | 13.0 | 17.0 |
| Vietnam | 9.7    | 6.0 | 9.0    | 13.0 | 17.0 |

| Country     | Mean ↓ | P25 | Median | P75  | P90  |
|-------------|--------|-----|--------|------|------|
| Singapore   | 9.0    | 5.0 | 8.0    | 12.0 | 17.0 |
| Taiwan      | 8.7    | 2.0 | 7.0    | 12.0 | 20.0 |
| Philippines | 8.2    | 3.0 | 7.0    | 11.0 | 16.0 |
| Malaysia    | 7.9    | 2.0 | 7.0    | 12.0 | 17.0 |
| India       | 7.6    | 3.0 | 6.0    | 11.0 | 16.0 |
| Mexico      | 6.3    | 3.0 | 5.0    | 9.0  | 13.0 |
| Thailand    | 5.8    | 1.0 | 4.0    | 9.0  | 14.0 |
| Indonesia   | 5.5    | 1.0 | 3.0    | 8.0  | 14.0 |
| Japan       | 5.5    | 2.0 | 5.0    | 8.0  | 11.0 |
| S. Korea    | 3.3    | 1.0 | 2.0    | 4.0  | 9.0  |
| Overall     | 7.2    | 2.0 | 6.0    | 10.0 | 15.0 |

## Sectoral demand structure (ISIC/NACE)

### Sectoral heatmap

Each job posting is assigned an economic sector via the ISIC/NACE classification derived from the ESCO occupation mapping (see [Step 06](#) above). At the 2-digit division level used in this analysis, NACE Rev. 2.1 and ISIC Rev. 5 are identical. The heatmap below shows the share of postings in each ISIC/NACE section (1-digit) per country. Clicking on any section row expands a 2-digit breakdown showing how demand is distributed across divisions within that sector.

| East Asia         |                                      | Benchmarks |      |      |      |      |      |      |      |      |
|-------------------|--------------------------------------|------------|------|------|------|------|------|------|------|------|
| ISIC/NACE section |                                      | JP         | KR   | TW   | TH   | MY   | SG   | ID   | VN   | PH   |
| ►N                | Professional, scientific & technical | 10.0       | 13.8 | 21.0 | 20.6 | 25.3 | 33.4 | 28.5 | 29.6 | 28.9 |
|                   | ►G Wholesale & retail trade          | 9.4        | 7.0  | 9.8  | 13.6 | 10.9 | 7.5  | 12.0 | 16.1 | 12.5 |
|                   | ►O Administrative & support          | 10.2       | 9.7  | 7.0  | 8.9  | 13.3 | 8.8  | 9.7  | 11.7 | 16.2 |
| ►K                | Telecom, IT & information services   | 13.3       | 5.5  | 17.1 | 2.8  | 4.4  | 10.8 | 6.5  | 7.5  | 5.3  |
|                   | ►I Accommodation & food service      | 7.8        | 4.4  | 5.4  | 23.6 | 15.0 | 7.1  | 14.8 | 5.0  | 5.2  |
|                   | ►C Manufacturing                     | 10.1       | 14.4 | 13.0 | 4.4  | 4.9  | 5.0  | 4.4  | 6.0  | 4.9  |
|                   | ►H Transportation & storage          | 8.4        | 4.3  | 4.7  | 8.7  | 5.4  | 5.0  | 4.5  | 4.2  | 6.4  |
|                   | ►R Human health & social work        | 9.1        | 12.4 | 2.6  | 1.1  | 2.8  | 2.6  | 1.3  | 1.3  | 3.0  |
|                   | Q Education                          | 3.8        | 5.8  | 2.6  | 1.9  | 2.9  | 4.3  | 3.3  | 3.1  | 2.1  |
|                   | ►S Arts, sports & recreation         | 2.6        | 2.4  | 3.3  | 4.1  | 3.9  | 2.3  | 3.7  | 2.1  | 2.0  |
|                   | ►L Financial & insurance             | 0.6        | 0.9  | 1.3  | 1.1  | 1.5  | 4.0  | 2.2  | 3.0  | 3.2  |
|                   | ►T Other service activities          | 2.5        | 1.8  | 1.9  | 2.9  | 2.6  | 2.3  | 2.5  | 1.9  | 2.1  |
|                   | ►F Construction                      | 3.5        | 2.4  | 2.9  | 1.0  | 0.9  | 1.5  | 0.8  | 1.1  | 0.9  |
| P                 | Public administration & defence      | 2.5        | 2.3  | 2.0  | 2.6  | 1.9  | 2.4  | 2.5  | 2.6  | 2.7  |
|                   | ►U Households as employers           | 2.5        | 7.9  | 0.3  | 0.5  | 0.4  | 0.1  | 0.2  | 0.2  | 0.5  |
|                   | ►J Publishing & broadcasting         | 0.8        | 1.1  | 2.5  | 0.8  | 1.9  | 1.1  | 1.3  | 2.0  | 1.2  |
|                   | M Real estate                        | 0.8        | 0.8  | 0.6  | 0.4  | 0.6  | 0.9  | 0.6  | 1.0  | 1.1  |
|                   | ►A Agriculture & fishing             | 0.6        | 1.6  | 0.9  | 0.5  | 0.7  | 0.4  | 0.7  | 1.0  | 0.9  |
|                   | D Electricity & gas                  | 0.4        | 0.4  | 0.5  | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  | 0.6  |
|                   | ►E Water supply & waste              | 0.7        | 0.7  | 0.2  | 0.1  | 0.2  | 0.1  | 0.2  | 0.2  | 0.3  |
|                   | ►B Mining & quarrying                | 0.3        | 0.3  | 0.4  | 0.1  | 0.2  | 0.1  | 0.2  | 0.1  | 0.1  |

JP Japan · KR S. Korea · TW Taiwan · TH Thailand · MY Malaysia · SG Singapore · ID Indonesia · VN Vietnam · PH Philippines

#### 💡 Statistical test: do countries differ in sectoral composition?

A global  $\chi^2$  test of independence on the 12×21 contingency table (country × ISIC/NACE section) yields  $\chi^2 = 413,673$ ,  $df = 220$ ,  $p < 0.001$ , confirming that sectoral demand structures differ significantly across the 12 countries. The overall effect size is Cramér's  $V = 0.14$ , indicating a moderate but meaningful departure from homogeneity.

**Pairwise similarity** (Cramér's  $V$ , lower = more similar):

| Most similar pairs    | $V$  | Most different pairs | $V$  |
|-----------------------|------|----------------------|------|
| Indonesia – Malaysia  | 0.11 | S. Korea – Thailand  | 0.43 |
| Vietnam – Philippines | 0.13 | Poland – Thailand    | 0.42 |

| Most similar pairs  | V    | Most different pairs | V    |
|---------------------|------|----------------------|------|
| Malaysia – Thailand | 0.16 | Indonesia – S. Korea | 0.41 |
| Singapore – Vietnam | 0.18 | S. Korea – Singapore | 0.40 |

## Macroeconomic and demographic context

The tables below provide a reference frame for interpreting the sectoral demand patterns. All data are from the [World Bank Open Data](#) (latest available year, 2023–2024), except Taiwan, for which data are sourced from the National Statistics Bureau and the National Development Council.

### OJS Error

SyntaxError: Unexpected end of JSON input

## Top occupation groups by sector

The table below shows the five most frequent ISCO-08 occupation groups (4-digit level) within each ISIC/NACE section, per country. Use the selectors to browse sectors and compare across countries.

**East Asia**   **Benchmarks**   **Sector:**

N Professional, Scientific And Technical Activities

### Japan JP · 30,121 postings

|      |                                            |       |
|------|--------------------------------------------|-------|
| 2149 | Engineering professionals not elsewhere... | 11.7% |
| 2431 | Advertising and marketing professionals    | 8%    |
| 1219 | Business services and administration ma... | 5.2%  |
| 7543 | Product graders and testers (excluding ... | 4.7%  |
| 3118 | Drafter                                    | 3.9%  |

### S. Korea KR · 23,375 postings

|      |                                            |       |
|------|--------------------------------------------|-------|
| 2431 | Advertising and marketing professionals    | 10.3% |
| 2149 | Engineering professionals not elsewhere... | 8%    |
| 2634 | Psychologists                              | 5.5%  |
| 3112 | Civil engineering technicians              | 3.9%  |
| 1221 | Sales and marketing managers               | 3.2%  |

### Taiwan TW · 24,741 postings

|      |                                            |       |
|------|--------------------------------------------|-------|
| 2149 | Engineering professionals not elsewhere... | 18.2% |
| 2166 | Graphic and multimedia designers           | 5.5%  |
| 2431 | Advertising and marketing professionals    | 5.5%  |
| 2152 | Integrated Circuit Design Engineer         | 5.3%  |
| 2141 | Packaging Production Manager               | 3.8%  |

### Thailand TH · 13,484 postings

|      |                                          |       |
|------|------------------------------------------|-------|
| 1221 | Sales and marketing managers             | 12.7% |
| 2431 | Advertising and marketing professionals  | 10.9% |
| 2411 | Accountants                              | 6.2%  |
| 2310 | University and higher education teachers | 6.2%  |
| 1211 | Finance managers                         | 5.3%  |

### Malaysia MY · 44,724 postings

|      |                                          |       |
|------|------------------------------------------|-------|
| 2431 | Advertising and marketing professionals  | 11.1% |
| 1221 | Sales and marketing managers             | 9.2%  |
| 1420 | Retail and wholesale trade managers      | 8.3%  |
| 2310 | University and higher education teachers | 5.5%  |
| 2411 | Accountants                              | 4.3%  |

### Singapore SG · 49,909 postings

|      |                                            |       |
|------|--------------------------------------------|-------|
| 2431 | Advertising and marketing professionals    | 10.4% |
| 2149 | Engineering professionals not elsewhere... | 7.6%  |
| 1221 | Sales and marketing managers               | 6.1%  |
| 2411 | Accountants                                | 5.7%  |
| 1420 | Retail and wholesale trade managers        | 4.8%  |

### Indonesia ID · 32,294 postings

|      |                                         |       |
|------|-----------------------------------------|-------|
| 2431 | Advertising and marketing professionals | 13.7% |
| 1221 | Sales and marketing managers            | 13.4% |
| 2411 | Accountants                             | 5.8%  |
| 1420 | Retail and wholesale trade managers     | 5.2%  |

### Vietnam VN · 39,797 postings

|      |                                         |       |
|------|-----------------------------------------|-------|
| 2431 | Advertising and marketing professionals | 14.3% |
| 2411 | Accountants                             | 12.7% |
| 3313 | Accounting associate professionals      | 10.1% |
| 1221 | Sales and marketing managers            | 7.2%  |

2643 Translators, interpreters and other lin...  
4.5%

4312 Statistical, finance and insurance cler...  
3.9%

#### Philippines PH - 50,724 postings

2431 Advertising and marketing professionals 14%  
3313 Accounting associate professionals 11.1%  
2411 Accountants 8.1%  
1221 Sales and marketing managers 6.5%  
1211 Finance managers 5.9%

### East Asia

#### Japan (300,124 postings)

2.3x R Health & social work 9.1%  
2.3x E Water & waste 0.7%  
2.2x U Households 2.5%  
0.3x L Financial & insurance 0.6%  
0.4x N Professional & scientific 10%

#### S. Korea (169,896 postings)

6.9x U Households 7.9%  
3.1x R Health & social work 12.4%  
2.5x E Water & waste 0.7%  
0.4x L Financial & insurance 0.9%  
0.5x I Accommodation & food 4.4%

#### Taiwan (117,665 postings)

2x K Telecom & IT 17.1%  
2x B Mining 0.4%  
1.8x F Construction 2.9%  
0.3x U Households 0.3%  
0.5x L Financial & insurance 1.3%

#### Thailand (65,541 postings)

2.8x I Accommodation & food 23.6%  
1.5x H Transport & storage 8.7%  
1.5x S Arts, sports & recreation 4.1%  
0.3x R Health & social work 1.1%  
0.3x K Telecom & IT 2.8%

#### Malaysia (176,653 postings)

1.8x I Accommodation & food 15%  
1.4x S Arts, sports & recreation 3.9%  
1.3x J Publishing & broadcasting 1.9%  
0.4x U Households 0.4%  
0.5x K Telecom & IT 4.4%

#### Singapore (149,538 postings)

1.6x L Financial & insurance 4%  
1.5x N Professional & scientific 33.4%  
1.3x K Telecom & IT 10.8%  
0.1x U Households 0.1%  
0.3x E Water & waste 0.1%

#### Indonesia (113,475 postings)

1.8x I Accommodation & food 14.8%  
1.3x S Arts, sports & recreation 3.7%  
1.2x N Professional & scientific 28.5%  
0.1x U Households 0.2%  
0.3x R Health & social work 1.3%

#### Vietnam (134,400 postings)

1.4x G Wholesale & retail 16.1%  
1.3x J Publishing & broadcasting 2%  
1.3x N Professional & scientific 29.6%  
0.2x U Households 0.2%  
0.3x R Health & social work 1.3%

#### Philippines (175,585 postings)

1.5x O Admin & support 16.2%  
1.4x M Real estate 1.1%  
1.4x D Utilities 0.6%  
0.4x U Households 0.5%  
0.5x B Mining 0.1%

### Benchmark countries

#### India (223,203 postings)

2.2x J Publishing & broadcasting 3.4%  
2.2x Q Education 7.8%  
1.6x M Real estate 1.2%  
0.2x U Households 0.2%  
0.4x H Transport & storage 2.4%

#### Mexico (219,745 postings)

2.2x L Financial & insurance 5.5%  
1.7x H Transport & storage 10.2%  
1.5x T Other services 3.6%  
0.3x J Publishing & broadcasting 0.5%  
0.3x U Households 0.4%

#### Poland (134,758 postings)

1.6x K Telecom & IT 13.2%  
1.5x G Wholesale & retail 17.6%  
1.4x L Financial & insurance 3.5%  
0.2x I Accommodation & food 1.3%  
0.4x U Households 0.5%

The cards above show each country's **most distinctive sectors** relative to the 12-country average. Green values indicate overrepresentation ( $> 1\times$  average), red values indicate underrepresentation ( $< 1\times$  average).

## Skill composition

### Cross-country comparison

The heatmaps below compare the composition of mapped ESCO concepts across **all countries at once**. Shares are normalised within each country separately for **skills**, **transversal skills**, and **knowledge**, so each column sums to 100% within the respective panel. The knowledge panel shows the most frequent 3-digit knowledge categories. Columns are labelled with ISO2 country codes and rows with short skill/knowledge codes — see the legends below each panel, and hover over any cell for the full labels and the exact share. Use the **descriptions** button to read the ESCO or ISCED-F definition behind each code. Language and other ESCO concept types are excluded from this specific view.

By default the panels show the **post-stratified** composition: each country's skill mix is reweighted so that occupation (ISCO-08) strata match its official employment structure, correcting for the over- or under-representation of occupations in online vacancies. Use the **Stratified / Raw** switch to compare with the unweighted sample, and the **method & sources** button for the full weighting methodology and the per-country reference data.

Stratified

Raw

Pooled

Within-posting

method & sources

## COUNTRIES

JP Japan KR South Korea TW Taiwan TH Thailand MY Malaysia SG Singapore ID Indonesia VN Vietnam PH Philippines IN India MX Mexico PL Poland

## Skills

Rows: default

Columns: default

descriptions

xlsx

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| S1 | 27.2%     | 21.5% | 28.3% | 27.3% | 26.0% | 29.5% | 28.7% | 30.7% | 27.1% | 27.5%     | 26.9% | 32.2% |
| S2 | 8.7%      | 10.7% | 12.0% | 12.1% | 13.1% | 14.3% | 12.3% | 13.6% | 14.4% | 15.3%     | 12.4% | 12.8% |
| S3 | 15.7%     | 15.2% | 12.4% | 15.7% | 15.0% | 11.5% | 15.4% | 10.4% | 12.2% | 10.7%     | 10.3% | 8.9%  |
| S4 | 27.1%     | 29.3% | 29.0% | 28.6% | 28.7% | 28.5% | 24.9% | 28.1% | 27.4% | 27.0%     | 27.5% | 26.2% |
| S5 | 5.7%      | 5.4%  | 6.7%  | 5.0%  | 4.5%  | 6.6%  | 5.0%  | 4.8%  | 5.0%  | 5.8%      | 4.6%  | 6.9%  |
| S6 | 9.4%      | 10.0% | 6.4%  | 6.4%  | 7.4%  | 6.1%  | 8.6%  | 7.2%  | 8.7%  | 8.0%      | 11.2% | 7.2%  |
| S7 | 2.1%      | 1.5%  | 1.7%  | 1.5%  | 1.5%  | 1.2%  | 1.7%  | 1.8%  | 1.7%  | 1.7%      | 2.0%  | 1.5%  |
| S8 | 4.2%      | 6.3%  | 3.4%  | 3.4%  | 3.9%  | 2.3%  | 3.4%  | 3.5%  | 3.4%  | 4.0%      | 5.1%  | 4.3%  |

S1 communication, collaboration and creativity S2 information skills S3 assisting and caring S4 management skills S5 working with computers S6 handling and moving S7 constructing S8 working with machinery and specialised equipment

## Transversal

Rows: default

Columns: default

descriptions

xlsx

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| T1 | 4.1%      | 3.4%  | 4.5%  | 2.0%  | 24.7% | 3.0%  | 7.0%  | 9.9%  | 3.3%  | 8.2%      | 2.8%  | 12.2% |
| T2 | 6.6%      | 17.0% | 25.1% | 18.0% | 10.9% | 19.6% | 14.1% | 15.2% | 17.9% | 21.7%     | 15.9% | 13.7% |
| T3 | 49.4%     | 34.9% | 31.2% | 27.9% | 28.7% | 38.6% | 27.0% | 33.8% | 33.7% | 31.7%     | 36.4% | 33.4% |
| T4 | 31.3%     | 29.3% | 30.3% | 32.3% | 25.2% | 29.5% | 33.7% | 33.5% | 31.4% | 29.0%     | 30.7% | 32.3% |
| T5 | 2.5%      | 4.1%  | 2.1%  | 9.3%  | 3.4%  | 2.1%  | 8.6%  | 2.5%  | 3.7%  | 2.9%      | 2.0%  | 0.9%  |
| T6 | 6.1%      | 11.4% | 6.8%  | 10.4% | 7.1%  | 7.2%  | 9.6%  | 5.1%  | 10.0% | 6.5%      | 12.1% | 7.5%  |

T1 core skills and competences T2 thinking skills and competences T3 self-management skills and competences T4 social and communication skills and competences T5 physical and manual skills and competences T6 life skills and competences

## Knowledge

Rows: default

Columns: default

descriptions

xlsx

|     | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|-----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|     | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| 041 | 22.6%     | 23.7% | 28.0% | 35.3% | 37.4% | 36.2% | 34.4% | 33.5% | 39.6% | 31.7%     | 36.4% | 28.5% |
| 071 | 6.6%      | 15.6% | 12.0% | 8.7%  | 9.7%  | 9.5%  | 9.4%  | 9.8%  | 6.6%  | 10.1%     | 9.8%  | 9.4%  |
| 061 | 9.4%      | 7.2%  | 16.0% | 5.7%  | 9.3%  | 12.9% | 7.5%  | 8.9%  | 7.7%  | 10.8%     | 6.3%  | 12.4% |
| 003 | 3.0%      | 2.9%  | 2.8%  | 4.9%  | 7.2%  | 6.2%  | 8.3%  | 5.1%  | 9.7%  | 7.7%      | 5.6%  | 5.5%  |
| 072 | 4.5%      | 7.6%  | 4.3%  | 3.7%  | 4.8%  | 3.5%  | 4.7%  | 4.4%  | 3.2%  | 4.7%      | 5.0%  | 2.6%  |
| 104 | 10.4%     | 7.0%  | 4.1%  | 2.2%  | 4.9%  | 2.5%  | 3.8%  | 1.0%  | 4.8%  | 2.5%      | 4.6%  | 4.5%  |
| 042 | 2.4%      | 2.2%  | 7.1%  | 8.5%  | 2.7%  | 2.7%  | 2.2%  | 7.8%  | 2.3%  | 1.9%      | 3.3%  | 8.7%  |
| 031 | 7.5%      | 1.7%  | 2.2%  | 8.5%  | 1.3%  | 1.6%  | 3.0%  | 3.8%  | 1.8%  | 0.9%      | 2.3%  | 1.9%  |
| 073 | 4.2%      | 4.9%  | 1.6%  | 1.9%  | 2.1%  | 2.8%  | 3.0%  | 3.2%  | 2.8%  | 3.3%      | 3.0%  | 3.2%  |
| 091 | 4.4%      | 3.0%  | 3.2%  | 1.8%  | 2.4%  | 3.0%  | 1.9%  | 1.5%  | 2.8%  | 3.0%      | 2.9%  | 4.7%  |
| 101 | 8.1%      | 1.6%  | 2.8%  | 1.7%  | 3.6%  | 2.6%  | 3.0%  | 2.6%  | 2.7%  | 1.7%      | 2.0%  | 1.0%  |
| 021 | 1.3%      | 1.4%  | 3.4%  | 1.8%  | 2.2%  | 2.0%  | 2.1%  | 2.1%  | 1.9%  | 3.4%      | 2.5%  | 1.4%  |

041 business and administration 071 engineering and engineering trades 061 information and communication technologies (icts) 003 personal skills and development 072 manufacturing and processing 104 transport services 042 law 031 social and behavioural sciences 073 architecture and construction 091 health 101 personal services 021 arts

## Skill Distribution By ISCO Group

## High-, Medium-, And Low-Skilled Occupations

The panels below split ESCO-mapped **skills** and **transversal skills** by broad occupation group: **high-skilled** occupations (ISCO 1-3), **medium-skilled** occupations (ISCO 4-8), and **low-skilled** occupations (ISCO 9). Each occupation group is shown as a separate full-width card, with shares normalised separately for each country and group. Use the **Pooled / Within-posting** switch to choose the estimator, and the **about shares** button for what each one means.

## COUNTRIES

JP Japan KR South Korea TW Taiwan TH Thailand MY Malaysia SG Singapore ID Indonesia VN Vietnam PH Philippines IN India MX Mexico PL Poland

Pooled

Within-posting

about shares

## Skill Distribution

descriptions

xlsx

## High-skilled occupations (ISCO 1-3)

Pooled mentions; each country column sums to 100% within this occupation group.

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| S1 | 35.0%     | 28.5% | 31.1% | 31.4% | 30.0% | 33.4% | 33.5% | 35.0% | 30.3% | 36.3%     | 30.5% | 35.1% |
| S2 | 9.8%      | 13.3% | 13.0% | 14.1% | 15.4% | 15.3% | 15.2% | 16.6% | 17.6% | 15.6%     | 14.7% | 14.9% |
| S3 | 11.2%     | 13.4% | 9.0%  | 15.2% | 11.1% | 8.7%  | 10.4% | 7.4%  | 8.6%  | 9.2%      | 8.9%  | 7.1%  |
| S4 | 27.7%     | 28.5% | 26.9% | 27.8% | 32.1% | 30.3% | 29.0% | 30.5% | 32.1% | 25.2%     | 31.2% | 27.8% |
| S5 | 9.3%      | 7.9%  | 11.2% | 6.4%  | 6.8%  | 8.3%  | 8.3%  | 7.1%  | 8.2%  | 9.6%      | 6.6%  | 10.2% |
| S6 | 3.7%      | 3.6%  | 4.1%  | 2.8%  | 2.5%  | 1.7%  | 1.6%  | 1.9%  | 1.6%  | 2.0%      | 4.2%  | 2.2%  |
| S7 | 1.0%      | 1.1%  | 1.8%  | 0.7%  | 0.6%  | 0.8%  | 0.7%  | 0.5%  | 0.5%  | 0.5%      | 1.0%  | 0.8%  |
| S8 | 2.2%      | 3.6%  | 2.9%  | 1.6%  | 1.7%  | 1.4%  | 1.3%  | 1.1%  | 1.1%  | 1.5%      | 2.8%  | 2.0%  |

## Medium-skilled occupations (ISCO 4-8)

Pooled mentions; each country column sums to 100% within this occupation group.

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| S1 | 26.7%     | 20.0% | 28.1% | 28.5% | 27.0% | 27.3% | 28.8% | 32.6% | 29.3% | 31.0%     | 28.5% | 34.2% |
| S2 | 7.8%      | 10.1% | 11.3% | 11.0% | 13.3% | 12.6% | 12.4% | 14.7% | 15.1% | 15.2%     | 12.4% | 11.1% |
| S3 | 17.0%     | 17.2% | 13.5% | 24.1% | 16.5% | 16.5% | 23.7% | 12.8% | 13.1% | 15.9%     | 11.1% | 9.3%  |
| S4 | 26.3%     | 28.7% | 29.1% | 21.7% | 28.3% | 26.5% | 21.3% | 27.8% | 28.1% | 23.1%     | 28.4% | 23.7% |
| S5 | 3.4%      | 4.8%  | 5.7%  | 4.0%  | 4.1%  | 4.4%  | 3.9%  | 4.4%  | 5.3%  | 6.0%      | 4.4%  | 4.9%  |
| S6 | 11.0%     | 8.5%  | 7.2%  | 7.0%  | 6.4%  | 7.6%  | 6.2%  | 4.4%  | 5.1%  | 5.1%      | 8.3%  | 10.9% |
| S7 | 2.0%      | 2.1%  | 1.6%  | 1.2%  | 1.2%  | 1.8%  | 1.1%  | 1.1%  | 1.2%  | 1.1%      | 1.8%  | 1.3%  |
| S8 | 5.9%      | 8.7%  | 3.5%  | 2.6%  | 3.2%  | 3.4%  | 2.6%  | 2.2%  | 2.7%  | 2.8%      | 5.0%  | 4.5%  |

## Low-skilled occupations (ISCO 9)

Pooled mentions; each country column sums to 100% within this occupation group.

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| S1 | 20.9%     | 14.5% | 26.3% | 20.9% | 19.2% | 14.3% | 20.3% | 23.3% | 18.3% | 21.0%     | 20.4% | 21.6% |
| S2 | 5.5%      | 6.6%  | 11.4% | 9.9%  | 9.7%  | 9.0%  | 9.2%  | 10.8% | 11.7% | 10.2%     | 10.3% | 8.4%  |
| S3 | 16.6%     | 11.3% | 13.0% | 25.1% | 18.0% | 17.2% | 22.0% | 15.5% | 15.2% | 18.1%     | 10.1% | 8.4%  |
| S4 | 25.5%     | 28.0% | 30.3% | 23.0% | 27.4% | 22.7% | 19.1% | 26.8% | 24.8% | 22.2%     | 27.4% | 27.6% |
| S5 | 1.4%      | 2.8%  | 5.5%  | 4.2%  | 2.3%  | 1.3%  | 3.1%  | 2.8%  | 2.7%  | 3.2%      | 3.0%  | 3.5%  |
| S6 | 24.0%     | 29.0% | 9.0%  | 13.2% | 17.2% | 30.1% | 21.5% | 16.9% | 21.7% | 21.2%     | 21.7% | 19.9% |
| S7 | 1.6%      | 1.7%  | 1.2%  | 1.1%  | 1.7%  | 1.2%  | 1.7%  | 1.1%  | 1.3%  | 0.9%      | 1.6%  | 1.2%  |
| S8 | 4.5%      | 6.0%  | 3.4%  | 2.7%  | 4.5%  | 4.2%  | 3.1%  | 2.8%  | 4.3%  | 3.2%      | 5.6%  | 9.3%  |

S1 communication, collaboration and creativity S2 information skills S3 assisting and caring S4 management skills S5 working with computers S6 handling and moving S7 constructing  
S8 working with machinery and specialised equipment

## Transversal distribution

descriptions

xlsx

## High-skilled occupations (ISCO 1-3)

Pooled mentions; each country column sums to 100% within this occupation group.

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| T1 | 6.0%      | 4.9%  | 5.9%  | 3.7%  | 20.0% | 3.7%  | 9.8%  | 13.4% | 3.4%  | 8.3%      | 3.7%  | 13.4% |
| T2 | 13.4%     | 23.3% | 27.6% | 23.6% | 16.6% | 25.5% | 24.4% | 18.9% | 26.4% | 25.8%     | 20.8% | 22.1% |
| T3 | 41.4%     | 31.5% | 29.4% | 25.8% | 31.3% | 35.3% | 28.2% | 30.7% | 32.9% | 30.7%     | 32.6% | 30.1% |
| T4 | 33.5%     | 27.1% | 30.4% | 31.7% | 25.0% | 28.1% | 29.0% | 31.7% | 29.2% | 26.9%     | 31.7% | 26.8% |
| T5 | 0.7%      | 2.3%  | 1.4%  | 7.1%  | 1.8%  | 0.5%  | 2.4%  | 0.9%  | 0.7%  | 1.8%      | 0.7%  | 0.4%  |
| T6 | 5.1%      | 10.9% | 5.3%  | 8.1%  | 5.4%  | 6.9%  | 6.2%  | 4.4%  | 7.5%  | 6.5%      | 10.5% | 7.2%  |

### Medium-skilled occupations (ISCO 4-8)

Pooled mentions; each country column sums to 100% within this occupation group.

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| T1 | 1.9%      | 2.8%  | 4.3%  | 2.0%  | 25.5% | 2.4%  | 6.0%  | 10.3% | 3.6%  | 8.2%      | 2.6%  | 11.2% |
| T2 | 5.5%      | 14.9% | 25.5% | 14.0% | 9.1%  | 13.6% | 11.9% | 14.4% | 19.4% | 18.4%     | 15.8% | 9.1%  |
| T3 | 52.3%     | 37.1% | 30.6% | 23.7% | 28.4% | 42.0% | 26.8% | 33.1% | 34.4% | 32.1%     | 35.8% | 32.9% |
| T4 | 31.2%     | 29.5% | 30.6% | 37.6% | 26.7% | 32.3% | 37.3% | 34.8% | 33.3% | 29.8%     | 32.5% | 37.8% |
| T5 | 3.4%      | 5.0%  | 2.2%  | 15.4% | 4.0%  | 3.3%  | 11.4% | 2.8%  | 1.8%  | 5.2%      | 1.8%  | 0.7%  |
| T6 | 5.8%      | 10.7% | 6.9%  | 7.3%  | 6.3%  | 6.5%  | 6.6%  | 4.6%  | 7.5%  | 6.3%      | 11.5% | 8.4%  |

### Low-skilled occupations (ISCO 9)

Pooled mentions; each country column sums to 100% within this occupation group.

|    | EAST ASIA |       |       |       |       |       |       |       |       | BENCHMARK |       |       |
|----|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|-------|-------|
|    | JP        | KR    | TW    | TH    | MY    | SG    | ID    | VN    | PH    | IN        | MX    | PL    |
| T1 | 1.0%      | 2.9%  | 4.1%  | 1.3%  | 27.6% | 2.0%  | 5.6%  | 7.5%  | 3.3%  | 4.1%      | 2.3%  | 8.2%  |
| T2 | 6.5%      | 13.5% | 24.2% | 7.9%  | 5.7%  | 4.7%  | 6.6%  | 8.7%  | 9.8%  | 11.6%     | 12.5% | 5.5%  |
| T3 | 56.0%     | 31.6% | 32.6% | 19.4% | 27.9% | 47.5% | 26.4% | 29.7% | 34.1% | 27.9%     | 38.1% | 42.0% |
| T4 | 26.1%     | 30.5% | 29.4% | 44.8% | 23.8% | 27.1% | 33.2% | 39.6% | 33.5% | 33.8%     | 28.7% | 34.6% |
| T5 | 3.4%      | 5.1%  | 2.4%  | 15.5% | 4.6%  | 8.7%  | 18.2% | 6.5%  | 7.6%  | 12.2%     | 3.5%  | 2.4%  |
| T6 | 7.0%      | 16.3% | 7.4%  | 11.2% | 10.4% | 10.1% | 10.1% | 8.1%  | 11.7% | 10.3%     | 15.0% | 7.3%  |

T1 core skills and competences    T2 thinking skills and competences    T3 self-management skills and competences    T4 social and communication skills and competences  
T5 physical and manual skills and competences    T6 life skills and competences

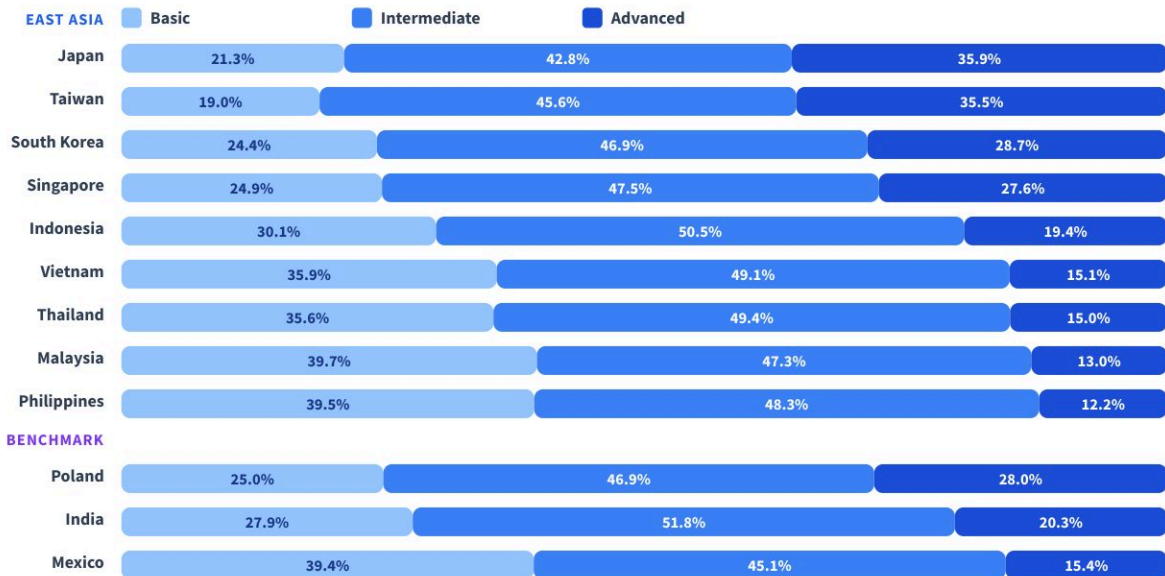
## Digital skills

### Digital skills prevalence

Using an adapted version of the digital-skill classification from [Chenin & Montorsi \(2025\)](#), the chart below shows the current **composition of digital skill mentions** after deduplication and the ICT-offer merge. Within each country, Basic, Intermediate, and Advanced sum to 100% of classified digital-skill mentions.

[xlsx](#)

Shares of classified digital-skill mentions; bars sum to 100% within each country.



## Digital skill mix over time

The previous section presents the composition of digital-skill demand as a single snapshot. This section examines how the mix changes over time by tracking the monthly distribution of digital skill mentions across three proficiency bands: Basic, Intermediate, and Advanced. To reduce short-term volatility, the series are presented as 3-month rolling averages. Within each country-month, the three bands sum to 100 percent.

The analysis is based on the composition of digital skill mentions rather than on the share of job postings that include at least one skill from a given proficiency band. This approach provides a more consistent measure of skill demand, as it focuses on the relative structure of extracted digital skills within each country and month. It also reduces sensitivity to differences in the number of skills extracted per posting, which may vary across data sources, occupations, posting formats, and processing conditions.

The monthly series are not directly comparable to the aggregate statistics presented above. For Japan, Korea, and Taiwan, a subset of postings from job portals contained only year-level information in 2026 and could therefore not be assigned to a specific month. These postings are included in the aggregate statistics but excluded from the monthly time-series analysis. As a result, the monthly estimates for these countries are based on a narrower subset of postings and may show a different skill composition. In particular, the aggregate data indicate a higher share of Advanced digital skills than is visible in the monthly series.

### MARKET

[xlsx](#)

#### EAST ASIA

**Indonesia**

Japan

Malaysia

Philippines

Singapore

South Korea

Taiwan

Thailand

Vietnam

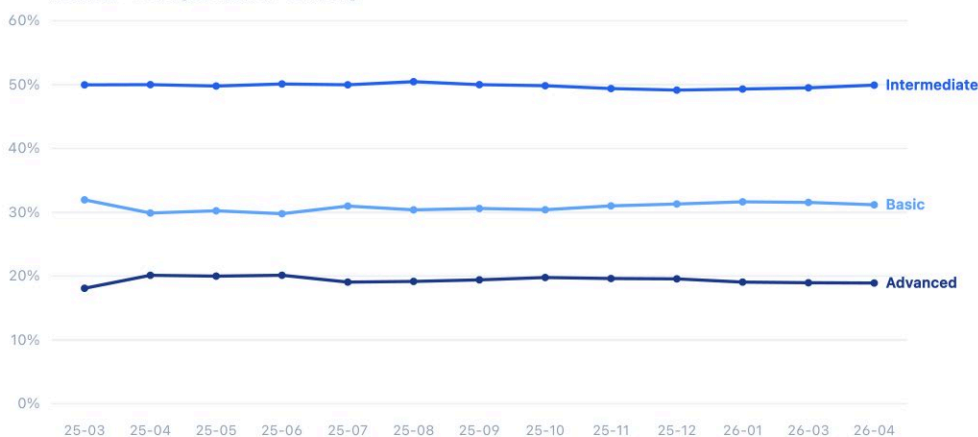
#### BENCHMARK

India

Mexico

Poland

Indonesia - % of digital mentions - 3-mo avg



3-month trailing rolling average of the digital-skill mention mix (the three bands sum to 100% within each country-month). Composition conditions on a digital mention, so it is not distorted by how many skills are extracted per posting.

## Advanced and intermediate digital profiles

The table below summarises each country's digital profile. Use the toggles to switch between the **Advanced** and **Intermediate** layers, between the two ranking methods, and between the **Limited** and **Wide** skill classifications. For each country it separates the most frequently mentioned concepts (**Common**) from those that are statistically over-represented relative to the rest of the sample (**Distinctive**); distinctive labels exclude any concept already shown as common.

The digital-skill classification is supplied externally (World Bank) and is offered in two variants. **Wide** uses the classification exactly as received. **Limited** is a cleaned view that resets a small set of false positives — concepts without genuine IT content (e.g. domain engineering, finance, teaching, or logistics labels mistakenly tagged as digital) — to non-digital, with totals and baselines recomputed accordingly. Limited is the preferred view; Wide is kept for transparency.

### Identifying distinctive digital skills by country

To identify which digital skills characterise each country's labour market, we draw on methods originally developed for computational text analysis — specifically, keyness analysis. The core intuition is straightforward: a skill is “distinctive” for a country not simply because it appears frequently there, but because it appears more often than one would expect given how common that skill is across all countries combined. This controls for the fact that some skills (e.g. communication tools) are universally demanded, while others are genuinely country-specific signals.

Concretely, we treat each country's pool of job postings as a corpus and each digital skill as a vocabulary item. For every skill, we compare its relative frequency in a given country against its relative frequency in the remaining eight countries pooled together. A skill that is over-represented in one country relative to this baseline is flagged as distinctive for that country; a skill that is under-represented is flagged as comparatively absent.

We apply two complementary measures. The first, the log-likelihood ratio ( $G^2$ , Dunning 1993), is a standard statistical test of whether the observed difference in frequency could plausibly arise by chance. The second, the weighted log-odds ratio (Monroe, Colaresi & Quinn 2008), goes a step further: it accounts for the fact that estimates based on rare skills are inherently less reliable than those based on common ones. By incorporating a smoothing prior and standardising by the uncertainty of each estimate, this approach prevents skills that happen to appear in only a handful of postings from dominating the rankings. We use the log-odds measure as our primary ranking criterion and  $G^2$  as a robustness check; results are consistent across both.

The analysis is conducted separately for intermediate and advanced digital skills, with each level's ranking computed relative to its own baseline frequency — ensuring that the two tiers are not conflated. The output for each country is a ranked list of skills that are distinctively over-represented, providing a compact, interpretable signature of each country's digital skill demand profile.

LAYER **Advanced** Intermediate METHOD **Log-odds**  $G^2$  CLASSIFICATION **Limited** Wide [xlsx](#)

**example** = most frequent within the country (Common) **example** = statistically over-represented vs. the rest of the sample (Distinctive)

| Country     | Characteristic labels                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Japan       | <b>web programming</b> <b>cloud technologies</b> <b>Java (computer programming)</b><br><b>model based system engineering</b> <b>systems development life-cycle</b>   |
| South Korea | <b>electronics</b> <b>electrical engineering</b> <b>civil engineering</b> <b>design buildings</b> <b>CAM software</b>                                                |
| Singapore   | <b>computer science</b> <b>cloud technologies</b> <b>computer engineering</b> <b>data analytics</b> <b>use scripting programming</b>                                 |
| Taiwan      | <b>electrical engineering</b> <b>cloud technologies</b> <b>computer engineering</b> <b>firmware</b> <b>hardware architectures</b>                                    |
| Malaysia    | <b>computer science</b> <b>perform data analysis</b> <b>electrical engineering</b> <b>SAP Data Services</b> <b>SQL</b>                                               |
| Thailand    | <b>computer science</b> <b>perform data analysis</b> <b>electrical engineering</b><br><b>use functional programming</b> <b>develop innovative mobility solutions</b> |
| Indonesia   | <b>computer science</b> <b>perform data analysis</b> <b>electrical engineering</b> <b>industrial engineering</b> <b>mining engineering</b>                           |
| Vietnam     | <b>perform data analysis</b> <b>computer science</b> <b>mechanical engineering</b> <b>use software design patterns</b> <b>Agile development</b>                      |
| Philippines | <b>perform data analysis</b> <b>computer science</b> <b>automation technology</b> <b>develop reporting software</b> <b>manage processes</b>                          |

Advanced digital profiles across East Asia (Limited classification), ranked by Log-odds (Monroe). Each country row separates the most common concepts from the most distinctive ones relative to the rest of the sample.

Most distinctive advanced digital terms [xlsx](#)

Ranked by Log-odds (Monroe) (Limited classification). For each country, bars show the digital concepts most over-used relative to the rest of the sample.

| Japan                                  | South Korea              |
|----------------------------------------|--------------------------|
| web programming z 115.09               | electronics z 129.63     |
| model based system engineering z 92.22 | design buildings z 72.12 |

|                                |         |
|--------------------------------|---------|
| systems development life-cycle | z 79.55 |
| Java (computer programming)    | z 77    |
| develop with cloud services    | z 71.46 |
| TypeScript                     | z 67.99 |

|                         |         |
|-------------------------|---------|
| electrical engineering  | z 55.76 |
| civil engineering       | z 46.95 |
| CAM software            | z 42.23 |
| quality control systems | z 35.73 |

### Singapore

|                           |         |
|---------------------------|---------|
| computer science          | z 54.07 |
| data analytics            | z 30.54 |
| use scripting programming | z 30.31 |
| cyber security            | z 30.04 |
| aerospace engineering     | z 28.06 |
| DevOps                    | z 23.7  |

### Taiwan

|                                       |         |
|---------------------------------------|---------|
| firmware                              | z 62.08 |
| hardware architectures                | z 60.78 |
| design firmware                       | z 50.09 |
| principles of artificial intelligence | z 48.01 |
| design hardware                       | z 47.96 |
| semiconductors                        | z 43.36 |

### Malaysia

|                            |         |
|----------------------------|---------|
| computer science           | z 36.2  |
| SAP Data Services          | z 34.77 |
| SQL                        | z 28.2  |
| develop reporting software | z 26.12 |
| mechatronics               | z 24.46 |
| manage processes           | z 22.76 |

### Thailand

|                                       |         |
|---------------------------------------|---------|
| use functional programming            | z 21.2  |
| develop innovative mobility solutions | z 21.02 |
| manage keys for data protection       | z 18.5  |
| innovate in ICT                       | z 18    |
| perform data analysis                 | z 17.98 |
| electrical engineering                | z 17.04 |

### Indonesia

|                             |         |
|-----------------------------|---------|
| computer science            | z 35.06 |
| industrial engineering      | z 30.6  |
| mining engineering          | z 25.25 |
| analyse information systems | z 20.15 |
| teach computer science      | z 17.75 |
| innovate in ICT             | z 16.87 |

### Vietnam

|                                 |         |
|---------------------------------|---------|
| mechanical engineering          | z 38.32 |
| perform data analysis           | z 35.94 |
| use software design patterns    | z 31.22 |
| Agile development               | z 30.37 |
| use object-oriented programming | z 25.7  |
| levels of software testing      | z 25.51 |

### Philippines

|                            |         |
|----------------------------|---------|
| develop reporting software | z 68.45 |
| manage processes           | z 60.55 |
| perform data analysis      | z 59.44 |
| SAP Data Services          | z 54.44 |
| industrial engineering     | z 54.19 |
| ensure information privacy | z 40.92 |

## Digital skill co-occurrences

This section examines how digital skills cluster together within a country's job postings, and what broader competences a digital role tends to require. Two complementary views are provided.

**Digital pairings** takes the most frequently requested digital skills and measures the strength of association within each pair using normalised pointwise mutual information (NPMI). NPMI adjusts for the underlying frequency of each skill, so that a high score reflects genuine co-demand rather than the prevalence of either skill on its own. Darker cells mark pairings that occur more often than chance would predict. Each skill is labelled by ICT proficiency level — Basic, Intermediate, or Advanced — making clusters readily visible, such as groupings of advanced skills with one another or of basic skills with one another.

**Digital × skill areas** extends the analysis beyond other digital skills to the wider requirements of the role. For each broad ESCO area — skill groups S1–S8, transversal competences T1–T6, and knowledge fields — it reports the additional likelihood, in percentage points, that a digital posting features that area relative to an average posting, disaggregated by ICT level. An accompanying table lists the specific non-digital skills most over-represented alongside digital roles at each level.

*Methodology.* Co-occurrence is measured at the mention level, using the same ESCO mapping applied throughout this report. Digital skills are classified by ICT proficiency level (Basic, Intermediate, Advanced). A posting is considered to “feature” an area when it contains at least one non-digital skill belonging to that area; the resulting premium therefore reflects the surrounding role rather than the digital skills themselves.

❗ OJS Error

SyntaxError: Unexpected end of JSON input

❗ OJS Runtime Error

Unexpected end of JSON input

## Engineers and technicians demand

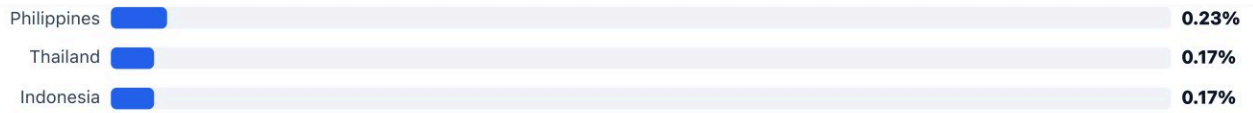
We use the ESCO occupation assigned to each posting title by JobBERT to identify vacancies for engineers and technicians. The chart can be switched between raw counts, postings per 100,000 labour force, and the share of all offers. Dashed reference lines show the average across East Asian markets and benchmark markets; click any country to inspect which specific ESCO occupations make up the selected category.



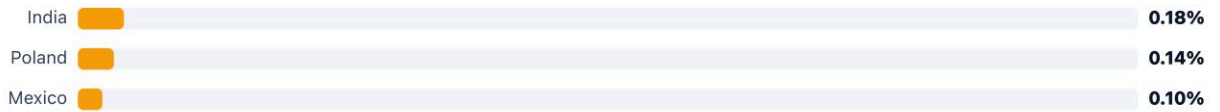
## Case study: semiconductors

Semiconductors are the strategic backbone of East Asia’s technology economy, which makes them a useful worked example of how the ESCO skill layer can isolate the demand signal for a single industry. We first identified every ESCO concept that explicitly refers to semiconductors or microelectronics — chip and integrated-circuit design, wafer fabrication and processing, micro-electro-mechanical systems (MEMS), and semiconductor inspection and testing — and then flagged any job posting that mentions at least one of them (ESCO match score  $\geq 0.50$ , de-duplicated by posting). The toggle below switches between the raw number of semiconductor postings, postings normalised per 100,000 people in the labour force, and the share of all postings in each market.





#### BENCHMARK



Share of all postings · 24 ESCO semiconductor concepts · match score  $\geq 0.5$

► The 24 ESCO concepts counted as "semiconductor"

Is semiconductor demand rising over time?

To test whether semiconductor demand is trending, we track the monthly share of postings that mention a semiconductor concept — semiconductor postings per 10,000 postings. The series runs from March 2025 to April 2026.

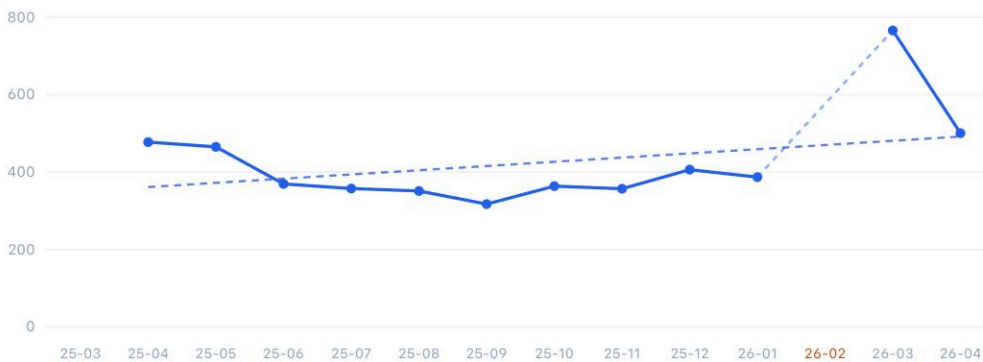
MARKET **EAST ASIA** **Taiwan** Japan South Korea Singapore Malaysia Thailand Indonesia Vietnam [xlsx](#)

Philippines

BENCHMARK

India Mexico Poland

per 10k postings



**Trend in semiconductor intensity (per 10k postings, reliable months only): upward and not statistically significant.** Mann-Kendall  $\tau = 0.12$  ( $p = 0.63$ ); OLS slope = +11.9 per 10k per month; mean  $\approx 426$  per 10k ( $\approx 4.26\%$  of postings); +26% comparing the first and last quarters.

The series now extends to April 2026. \* **February 2026** is shown for completeness only: owing to the timing of data collection that month is only partially covered, so its value is unreliable and is bridged with a dashed line. Trend statistics use reliable months only.

## Case study: artificial intelligence

We repeat the exercise for artificial intelligence, this time drawing on ESCO **skills** (e.g. *utilise machine learning, develop computer vision system, perform data mining*) and **knowledge** concepts (e.g. *principles of artificial intelligence, machine learning, deep learning, natural language processing*). The concepts are grouped into six families — general AI, machine learning & neural networks, natural language & speech, computer vision, data mining & predictive analytics, and robotics & automation — and the coloured dots throughout this section encode that family (see the legend). The toggle switches between the raw number of AI postings, postings per 100,000 labour force, and the share of all postings; click any country to open a breakdown of which AI concepts drive its demand.

METRIC **% of postings** per 100k LF count [xlsx](#)

Concept family: ● General AI ● Machine learning & neural networks ● Natural language & speech ● Computer vision ● Data mining & predictive analytics ● Robotics & automation

EAST ASIA





#### BENCHMARK

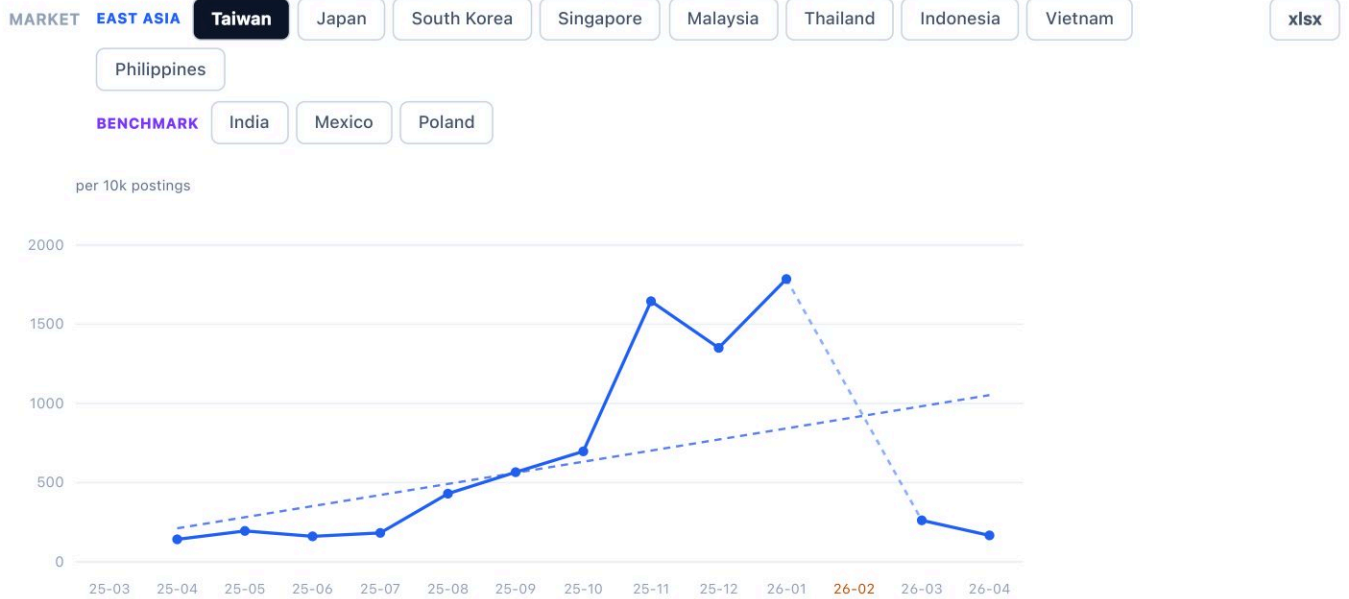


Share of all postings · 17 ESCO AI concepts (skills + knowledge) · match score  $\geq 0.5$  · click a country for the per-concept breakdown

► [The 17 ESCO AI concepts counted](#)

### Is AI demand rising over time?

As before, we track the monthly share of postings mentioning an AI concept, per 10,000 postings, from March 2025 to April 2026.



**Organic trend: upward, statistically significant.** Mann-Kendall  $\tau = 0.45$  ( $p = 0.05$ ); OLS slope = +76.5 per 10k/month; mean  $\approx 632$  per 10k ( $\approx 6.32\%$  of postings); +345% first vs. last quarter.

The series now extends to April 2026. \* **February 2026** is shown for completeness only: owing to the timing of data collection that month is only partially covered, so its value is unreliable and is bridged with a dashed line. Trend statistics use reliable months only.

## Technical Annex

Each processing step is described in detail below.

### STEP 01 Language detection

Comparing skill demand across countries that operate in different languages requires, as a first step, reliable identification of the language in which each posting was written. This ensures that postings already in English can proceed directly to analysis, while those in other languages are routed to the appropriate translation track.

We apply a machine-learning-based language detector ([Lingua ↗](#)) independently to both the job title and the full job description of every posting. The detector covers all languages present in our sample, including CJK scripts (Chinese, Japanese, Korean), Thai, Vietnamese, and Latin-script languages (English, Spanish, Malay, Filipino, Polish). Per-field detection allows us to handle the common case where a title is in English but the body is in the local language. This is important because mixed-language postings are common in online vacancies, and Lingua performs well even when different parts of the same offer use different languages.

### STEP 02 Translation

Cross-country skill comparison requires a common-language representation of all job postings. With a corpus of nearly **2 million job ads**, machine translation is not an optional enhancement but a practical necessity: without it, large-scale cross-country comparison would not be feasible in a

consistent and timely manner. Recent research indicates that LLM-based translation systems can deliver high-quality machine translation at scale, supporting the efficient processing of large datasets such as the one used in this study (see [Huang et al. \(2023\)](#) ).

Postings already detected as English pass through unchanged. All remaining postings are translated with [Gemma 3 Translate 4B](#) , an open source multilingual translation model designed for high-quality instruction following and strong performance across typologically different languages. Both **job titles** and **job descriptions** are translated into English and stored alongside the originals to allow back-referencing when analysts need to inspect the source text. This ensures that all downstream steps, including title standardisation, operate on a common-language representation.

To validate translation quality, we conducted a **back-translation** exercise on a small random sample of Polish postings. In this check, texts were translated into English and then translated back into Polish, allowing us to verify that the meaning of skills, occupations, and task descriptions was preserved with minimal drift.



Translation routing

### STEP 03 Skill-sentence filtering

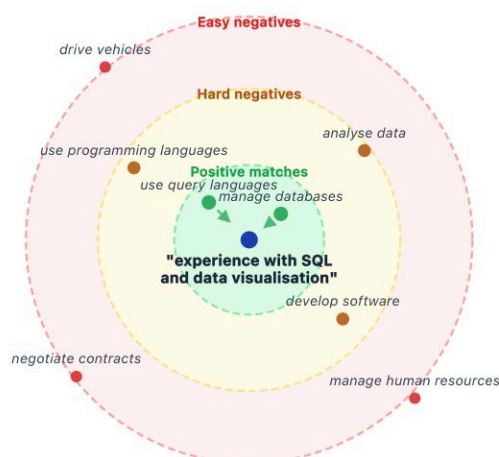
Job descriptions typically mix skill requirements with administrative content such as benefits, company history, or application instructions. Isolating the skill-relevant sentences before downstream mapping reduces noise and improves the precision of the extracted skill profiles.

We use a fine-tuned BERT classifier based on the [Malandri et al. \(2025\)](#) SkillLess approach and the [BERT-OJA-SkillLess](#) . The model performs binary sentence-level classification, labelling each sentence as either skill-relevant or non-skill. In our sample, approximately 40–60 % of description text is filtered out as non-skill content. Only the retained sentences proceed to the ESCO mapping step.

### STEP 04 ESCO skill mapping

The European Skills, Competences, Qualifications and Occupations ([ESCO](#) ) taxonomy provides a standardised vocabulary of roughly 13,000 skills and competences maintained by the European Commission. Mapping extracted skill sentences to ESCO entries creates a common analytical framework that enables direct comparison across countries and languages.

We employ a **custom contrastive bi-encoder** trained on a massive synthetic dataset comprising hundreds of thousands of ESCO-labelled examples derived from the ESCO classification structure. The bi-encoder learns a **shared representation space** for two different objects — job-ad sentences and standardised ESCO skill labels — by bringing semantically matching pairs closer together while pushing non-matching pairs apart. In practical terms, consider a job-ad sentence such as *“experience with SQL and data visualisation”*. The model learns to place it close to the correct ESCO entries — *“use query languages”* and *“manage databases”* — and away from everything else. Crucially, the training signal is strongest for **hard negatives**: skills that are semantically nearby but not a correct match, such as *“use programming languages”* or *“develop software”*. These are more informative than trivially distant skills like *“negotiate contracts”*, because the model must learn fine-grained distinctions within the same domain rather than merely separating unrelated concepts. During training, correct sentence–skill pairings act as positives while all other skills in the same batch act as negatives — with the hardest cases (same-domain, wrong skill) driving most of the learning signal. This gives the model a scalable way to learn ranking behaviour across the full ~13,000-skill taxonomy without a separate classifier for each skill.



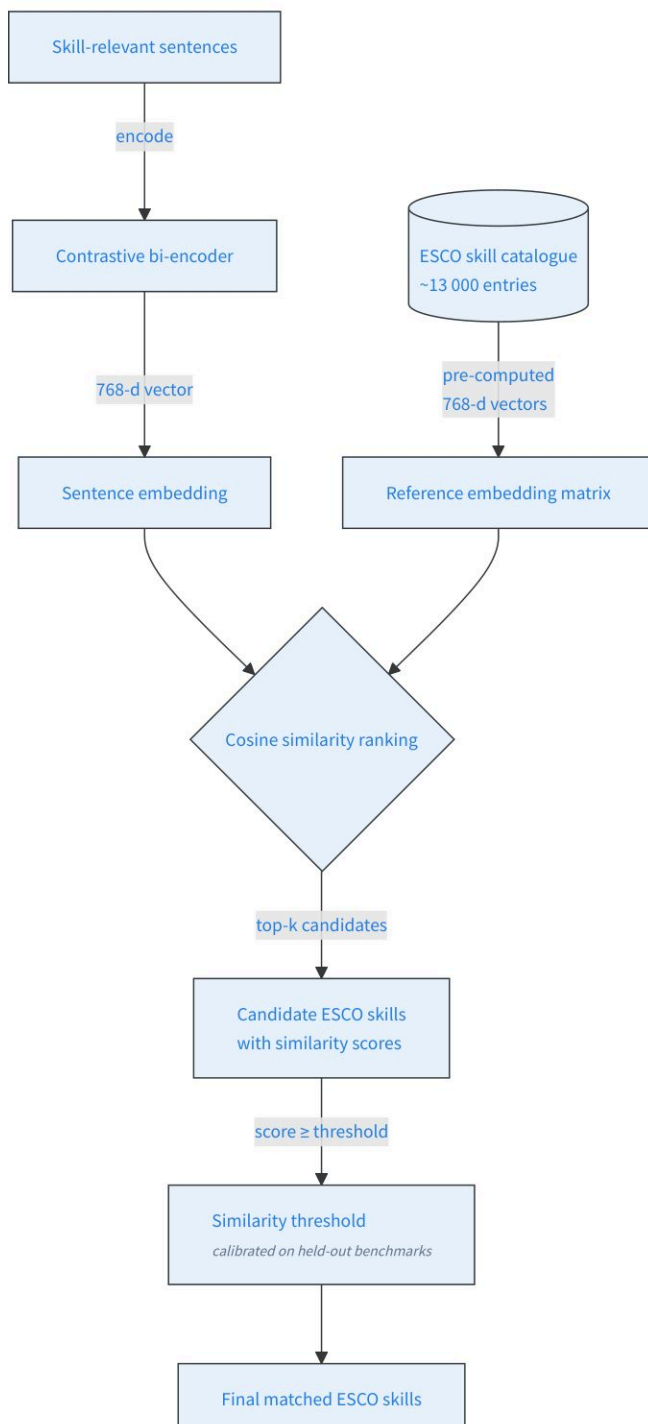
Simplified 2-D projection of the embedding space (actual representations are high-dimensional). Own elaboration.

This approach is well suited to online job ads because employers often describe the same skill in many different ways. By learning semantic proximity rather than relying on exact wording, the model can recover skills even when the phrasing differs from the ESCO label itself.

Our approach also avoids relying only on one averaged sentence representation. Instead, the encoder can attend more strongly to the parts of a sentence that are most diagnostic for a given skill, which is particularly useful when a single sentence mentions several tasks, tools, or competencies at once. This token-level attention mechanism has been shown to improve matching quality in multi-skill sentences ([Sun, 2026](#); [Decorte et al., 2025](#)).

**Similarity threshold calibration.** Not every high-ranking candidate is a true match. Following established practice in ESCO skill extraction ([Decorte et al., 2025](#); [Bielinski & Brazier, 2025](#)), we retain only those candidates whose cosine similarity exceeds a calibrated threshold. The threshold is tuned on held-out benchmark data (the same HOUSE and TECH datasets used to evaluate overall pipeline performance) by selecting the value that maximises the trade-off between precision and recall. This step filters out low-confidence matches and controls the total number of skills assigned per posting.

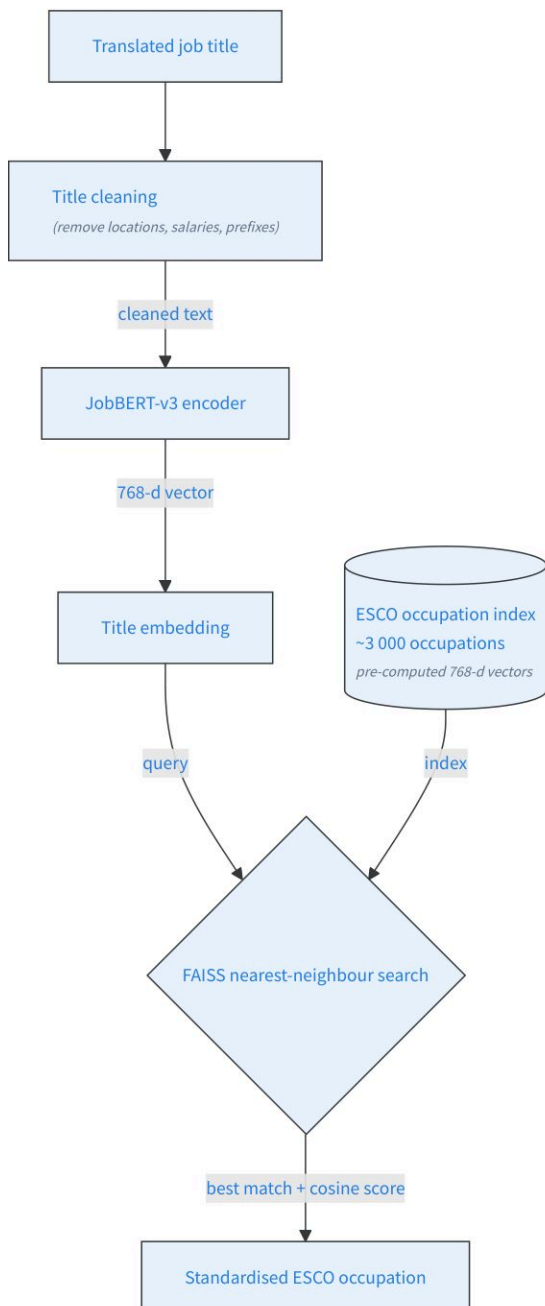
① Our skill-mapping pipeline outperforms the current state of the art by approximately 3 percentage points in RP@5 on the HOUSE and TECH benchmark datasets reported by [Decorte et al. \(2025\)](#). These benchmarks consist of job postings annotated with ESCO gold-standard skill labels by human experts, against which model predictions are evaluated. The reported RP@5 scores are 0.822 and 0.838, respectively.



**STEP 05 Job title standardisation**

Raw job titles are highly variable: the same role may appear as “Software Engineer”, “Backend Developer”, “Platform Engineer”, “Full-Stack Application Developer”, or “Senior API Engineer”. Likewise, an analytical role may appear as “Data Analyst”, “BI Analyst”, “Reporting Specialist”, or “Insights Associate”. Because titles are first translated into English, the standardisation step works on a harmonised title field rather than directly on multilingual raw text. Mapping each title to a standardised ESCO occupation enables comparable occupational analysis and serves as the bridge to sector classification in the next step.

Titles are first cleaned by removing common noise (embedded locations, salary ranges, seniority prefixes). The cleaned titles are then embedded using [JobBERT-v3](#), a domain-specific open model pre-trained on large-scale job market data. We pre-compute embeddings for all ~3,000 ESCO occupation labels and store them in a [FAISS](#) vector index for fast retrieval. Each job title is matched to the nearest ESCO occupation by cosine similarity, with the similarity score retained for downstream quality filtering.



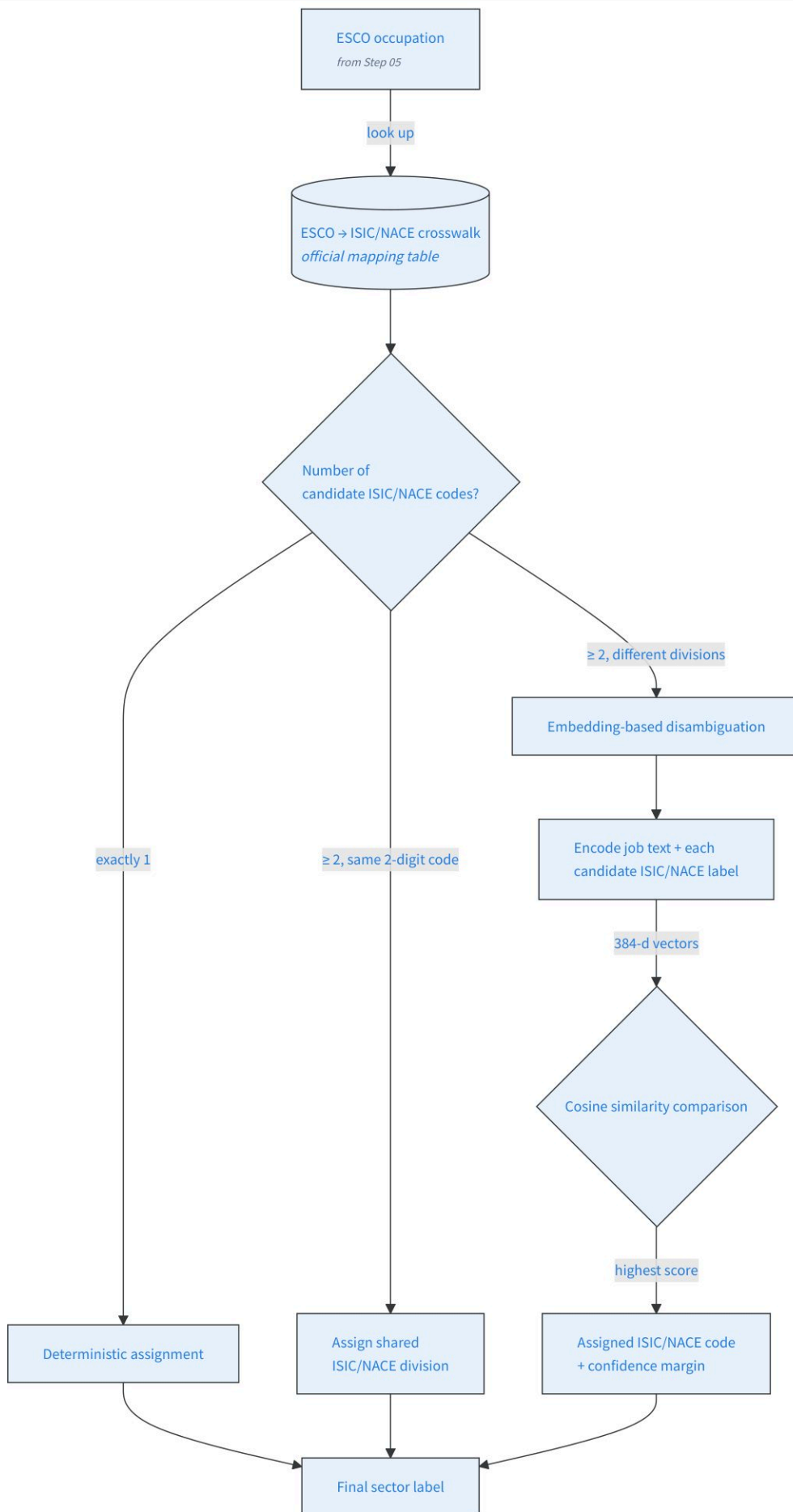
Job title standardisation

**STEP 06 Sector classification (ISIC/NACE)**

Assigning an economic sector to each job posting enables analysis of skill demand by industry—a dimension that is critical for targeted workforce development and industrial policy. We classify postings into the **ISIC/NACE** industry taxonomy. The underlying crosswalk uses the full NACE Rev. 2.1 classification maintained by Eurostat, which provides finer sub-division detail (3- and 4-digit codes) than ISIC. However, because this report analyses sectoral demand at the **2-digit division level**, the results are identical under both systems: at 2 digits, NACE Rev. 2.1 and ISIC Rev. 5 share the same division codes and boundaries ([Eurostat, 2024 ↗](#)). All results in this report are therefore directly comparable with any analysis using either framework.

The assignment relies on the official [ESCO-to-NACE crosswalk ↗](#), which links each ESCO occupation to one or more ISIC/NACE activity codes. Three resolution tiers handle different levels of ambiguity:

1. **One-to-one mapping** — the ESCO occupation maps to a single ISIC/NACE code. The assignment is deterministic.
2. **Same 2-digit code** — all candidate codes share the same 2-digit division (e.g. all fall under C26). The shared division is assigned directly, since finer sub-division differences are below the reporting threshold.
3. **Ambiguous multi-division cases** — candidates span different ISIC/NACE divisions. We resolve these by embedding the job title and description together with each candidate activity label using a multilingual sentence-transformer ([paraphrase-multilingual-MiniLM-L12-v2 ↗](#)) and selecting the candidate with the highest cosine similarity. A confidence margin between the top two candidates is recorded to support post-hoc quality auditing.



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